

03_orphan_records

Propietario	© Carlos Roberto
Verificación	Verificada
Etiquetas	Arquitectura de Datos DBA

3.1 FACTURA with NULL pedido_id

```
SELECT 'SECTION 1 - Type A: Invoices with NULL pedido_id (no
commercial origin)' AS diagnostic_section;

SELECT
    f.factura_id,
    f.numero_factura,
    f.cliente_fact_id,
    fc.nit_cliente,
    fc.razon_social,
    f.pedido_id,
    f.fecha_emision,
    f.fecha_vencimiento,
    f.total,
    f.estado_pago,
    f.observaciones,
    DATEDIFF(DAY, f.fecha_emision, GETDATE()) AS days_since_i
ssued,
    'NULL_PEDIDO_ID' AS orphan_type
FROM facturacion.FACTURA f
LEFT JOIN facturacion.CLIENTE fc ON f.cliente_fact_id = fc.cl
iente_fact_id
WHERE f.pedido_id IS NULL
ORDER BY f.total DESC;
```

diagnostic_section											
1	SECTION 1 - Type A: Invoices with NULL pedido_id...										
	factura_id	numero_factura	cliente_fact_id	nit_cliente	razon_social	pedido_id	fecha_emision	fecha_vencimiento	total	estado_pago	ob...
76	480672	FAC-ORPHAN-00672	978	978	MATERIALES DE CONSTRUCCION GT S.A...	NULL	2022-12-18 02:07:05.7700000	2023-08-11	55248.48	PAGADA	NL
77	481509	FAC-ORPHAN-01509	11723	11723	COMERCIAL PACIFICO, LTDA. - 11723	NULL	2022-04-18 02:07:06.4766667	2024-03-19	55217.12	PAGADA	NL
78	484210	FAC-ORPHAN-04210	10775	10775	PROCESADORA DE ALIMENTOS GT S.A. ...	NULL	2023-08-28 02:07:09.4033333	2024-10-26	55194.72	PENDIENTE	NL
79	483971	FAC-ORPHAN-03971	648	648	SUMINISTROS GENERALES, LTDA. - 648	NULL	2023-03-26 02:07:09.1100000	2024-01-21	55156.64	VENCIDA	NL
80	480212	FAC-ORPHAN-00212	8862	8862	IMPORTACIONES DEL NORTE S.A. - 8862	NULL	2024-07-29 02:07:05.3100000	2024-05-23	55139.84	VENCIDA	NL
81	484473	FAC-ORPHAN-04473	9629	9629	ALMACENADORA DEL ESTE S.A. - 9629	NULL	2024-08-21 02:07:09.7500000	2024-11-02	55127.52	PAGADA	NL
82	481476	FAC-ORPHAN-01476	4491	4491	CONSTRUCTORA REGIONAL GT S.A. - 44...	NULL	2024-09-12 02:07:06.4566667	2024-09-09	55114.08	PAGADA	NL
83	481634	FAC-ORPHAN-01634	4804	4804	EXPORTADORA CENTROAMERICANA S.A...	NULL	2023-05-11 02:07:06.5700000	2022-04-09	55091.68	VENCIDA	NL
84	483698	FAC-ORPHAN-03698	5145	5145	INDUSTRIAS UNIDAS, S.A. - 5145	NULL	2024-12-07 02:07:08.7866667	2024-07-15	55090.56	VENCIDA	NL
85	483453	FAC-ORPHAN-03453	3591	3591	CONSTRUCTORA REGIONAL GT S.A. - 35...	NULL	2025-07-27 02:07:08.5100000	2025-02-17	55073.76	PAGADA	NL
86	481784	FAC-ORPHAN-01784	2800	1301	DISTRIBUIDORA NACIONAL, S.A. - 2800	NULL	2024-02-23 02:07:06.6666667	2021-07-23	55069.28	VENCIDA	NL
87	481600	FAC-ORPHAN-01600	8649	8649	ALMACENADORA DEL ESTE S.A. - 8649	NULL	2025-04-05 02:07:06.5500000	2026-06-16	55063.68	PENDIENTE	NL
88	481256	FAC-ORPHAN-01256	5218	5218	MATERIALES DE CONSTRUCCION GT S.A...	NULL	2026-01-06 02:07:06.3000000	2025-08-16	55025.60	VENCIDA	NL
89	484590	FAC-ORPHAN-04590	3400	401	DISTRIBUIDORA NACIONAL, S.A. - 3400	NULL	2025-07-21 02:07:09.8800000	2022-02-06	55021.12	PAGADA	NL
90	482551	FAC-ORPHAN-02551	6180	6180	DISTRIBUIDORA NACIONAL, S.A. - 6180	NULL	2025-03-29 02:07:07.4700000	2023-03-30	54992.00	PENDIENTE	NL
91	483845	FAC-ORPHAN-03845	3295	3295	PROCESADORA DE ALIMENTOS GT S.A. ...	NULL	2022-07-18 02:07:08.9666667	2025-02-27	54988.64	VENCIDA	NL
92	484863	FAC-ORPHAN-04863	1046	1046	AGRO EXPORTACIONES GT S.A. - 1046	NULL	2021-08-24 02:07:10.1533333	2023-11-10	54987.52	PAGADA	NL
93	483959	FAC-ORPHAN-03959	2463	2463	COMERCIAL PACIFICO, LTDA. - 2463	NULL	2025-01-26 02:07:09.0966667	2021-10-20	54987.52	VENCIDA	NL
94	483555	FAC-ORPHAN-03555	5513	5513	COMERCIAL PACIFICO, LTDA. - 5513	NULL	2025-05-12 02:07:06.9566667	2025-11-20	54965.28	PENDIENTE	NL

Query executed successfully.

c13lab_db (16.0 RTM)

carlos_roberto (52)

TRANSTRACK

00:00:00

Row: 1, Col: 1

5,001 rows

These are invoices issued without any associated purchase order - there is no way to know what service was provided, for which customer operationally, or if the delivery took place.

Financial summary for type A

```
-- Financial summary for type A
SELECT
    COUNT(*) AS orphan_invoice_count,
    SUM(f.total) AS total_orphan_value,
    AVG(f.total) AS avg_orphan_value,
    MIN(f.fecha_emision) AS earliest_orphan,
    MAX(f.fecha_emision) AS latest_orphan,
    COUNT(DISTINCT f.cliente_fact_id) AS clients_with_orphan_invoices,
    SUM(CASE WHEN f.estado_pago = 'PAGADA' THEN f.total ELSE 0 END) AS already_collected,
    SUM(CASE WHEN f.estado_pago = 'PENDIENTE' THEN f.total ELSE 0 END) AS pending_collection
FROM facturacion.FACTURA f
WHERE f.pedido_id IS NULL;
```

	orphan_invoice_count	total_orphan_value	avg_orphan_value	earliest_orphan	latest_orphan	clients_with_orphan_invoices	already_collected	pending_collection
1	5000	141608042.24	28321.608448	2021-06-02 02:07:05.9566667	2026-05-31 02:07:09.6033333	4104	47495854.56	46977762.72

Query executed successfully. c13/lab_db (16.0 RTM) | carlos_roberto (52) | TRANSTRACK | 00:00:00 | Row: 1, Col: 1 | 1 rows

5,000 invoices totaling **\$141.6M** have no associated order. **\$47.5M already collected**, \$47M pending - none traceable to a commercial transaction. Direct SAT audit risk under Decreto 27-92.

3.2 PEDIDO with no ENTREGA

```
SELECT 'SECTION 2 - Type B: Orders with no delivery record' AS diagnostic_section;
```

```
SELECT
    p.pedido_id,
    p.numero_pedido,
    p.cliente_id,
    vc.nit AS client_nit,
    vc.nombre AS client_name,
    p.ruta_id,
    r.ciudad_origen,
    r.ciudad_destino,
    p.fecha_pedido,
    p.fecha_requerida,
    p.estado,
    p.peso_kg,
    p.tipo_carga,
    p.valor_declarado,
```

```

-- SLA breach: required date passed with no delivery
CASE
    WHEN p.fecha_requerida < CAST(GETDATE() AS DATE)
        AND p.estado NOT IN ('CANCELADO')
    THEN 'SLA_BREACHED'
    ELSE 'NO_SLA_BREACH_YET'
END
AS sla_status,
DATEDIFF(DAY, p.fecha_requerida, GETDATE()) AS days_past_
required
FROM operaciones.PEDIDO p
LEFT JOIN operaciones.ENTREGA e ON p.pedido_id = e.pedido_id
LEFT JOIN ventas.CLIENTE vc ON p.cliente_id = vc.cliente_id
LEFT JOIN operaciones.RUTA r ON p.ruta_id = r.ruta_id
WHERE e.entrega_id IS NULL -- No delivery re
cord
AND p.estado NOT IN ('CANCELADO') -- Not cancelled
ORDER BY p.valor_declarado DESC, p.fecha_requerida ASC

SELECT
    COUNT(*) AS orders_without_deliver
y,
    SUM(p.valor_declarado) AS declared_value_untrack
ed,
    SUM(CASE WHEN p.fecha_requerida < CAST(GETDATE() AS DATE)
THEN 1 ELSE 0 END) AS confirmed_sla_breaches
FROM operaciones.PEDIDO p
LEFT JOIN operaciones.ENTREGA e ON p.pedido_id = e.pedido_id
WHERE e.entrega_id IS NULL AND p.estado NOT IN ('CANCELADO');

```

diagnostic_section												
1	SECTION 2 - Type B: Orders with no delivery record											
	pedido_id	numero_pedido	cliente_id	client_nit	client_name	ruta_id	ciudad_origen	ciudad_destino	fecha_pedido	fecha_requerida	estado	p
1	282354	PED-000280949	12053	NIT-012053	Farmaceutica Central 12053	1869	Alta Verapaz	San Marcos	2026-03-25 00:00:00.0000000	2023-11-30	EN_RUTA	1
2	73730	PED-000072325	7654	NIT-007654	Textilera Guatemala 7654	689	Izabal	Chiquimula	2026-05-18 00:00:00.0000000	2024-05-21	ASIGNADO	9
3	348877	PED-000347472	7568	NIT-001569	Suministros Generales Ltda 7568	1492	Izabal	Chimaltenango	2026-03-31 00:00:00.0000000	2025-05-16	EN_RUTA	1
4	70821	PED-000069416	3521	NIT-003521	Grupo Logistico GT 3521	111	El Quiche	Jutiapa	2026-03-30 00:00:00.0000000	2019-12-31	PENDIENTE	2
5	106450	PED-000105045	12065	NIT-012065	Industrias Unidas S.A. 12065	2000	Mazatenango	Solola	2026-03-24 00:00:00.0000000	2024-08-25	ENTREGADO	2
6	306764	PED-000305359	6023	NIT-006023	Comercial Pacifico Ltda. 6023	1369	Chimaltenango	Sacatepequez	2026-02-13 00:00:00.0000000	2026-02-24	ASIGNADO	2
7	304847	PED-000303442	10325	NIT-010325	Industrias Unidas S.A. 10325	455	Puerto Barrios	Jutiapa	2026-02-21 00:00:00.0000000	2022-03-25	ASIGNADO	1
8	358297	PED-000356892	13610	NIT-013610	Frigorifico Nacional 13610	1696	Alta Verapaz	Zacapa	2026-04-24 00:00:00.0000000	2026-04-10	ENTREGADO	1
9	481922	PED-000480517	6704	NIT-000704	Exportadora Centroamericana ...	242	Sacatepequez	Solola	2026-04-07 00:00:00.0000000	2022-10-04	PENDIENTE	1
10	383702	PED-000382297	642	NIT-000642	Importaciones del Norte 642	985	Puerto Barrios	Coban	2026-02-21 00:00:00.0000000	2025-06-10	PENDIENTE	1
11	333189	PED-000331784	6495	NIT-006495	Procesadora de Alimentos GT ...	2315	Santa Rosa	Alta Verapaz	2026-04-11 00:00:00.0000000	2021-04-30	PENDIENTE	1
12	167869	PED-000166464	8188	NIT-008188	Suministros Generales Ltda 8188	2294	Jutiapa	Puerto Barrios	2026-03-28 00:00:00.0000000	2019-11-24	EN_RUTA	2
13	479531	PED-000478126	4381	NIT-004381	Grupo Logistico GT 4381	1432	Izabal	Santa Rosa	2026-02-27 00:00:00.0000000	2024-11-16	EN_RUTA	4
14	214998	PED-000213593	14927	NIT-014927	Corporacion Logistica SA 14927	999	El Quiche	Sacatepequez	2026-05-08 00:00:00.0000000	2026-01-07	ASIGNADO	1
15	291999	PED-000290594	10410	NIT-010410	Frigorifico Nacional 10410	1933	Jalapa	Mazatenango	2026-05-04 00:00:00.0000000	2022-12-14	ASIGNADO	2
16	456682	PED-000455277	6457	NIT-006457	Electrodomesticos del Sur 6457	245	Baja Verapaz	Peten	2026-03-21 00:00:00.0000000	2024-03-05	ENTREGADO	2
17	247922	PED-000246517	6280	NIT-000281	Distribuidora Nacional S.A. 6280	2002	Chimaltenango	Totonicapan	2026-05-21 00:00:00.0000000	2023-06-14	ENTREGADO	5
1												
	orders_without_delivery	declared_value_untracked		confirmed_sla_breaches								
1	11799	599661971.00		11761								

Query executed successfully.

c13/lab_db (16.0 RTM) | carlos_roberto (52) | TRANSTRACK | 00:00:35 | Row: 1, Col: 1 | 1 row

Query executed successfully.

c13/lab_db (16.0 RTM) | carlos_roberto (52) | TRANSTRACK | 00:00:35 | Row: 1, Col: 1 | 1 rows

11,799 orders have no delivery record, representing **\$599.6M** in untracked cargo value. Of these, **11,761 have a confirmed SLA breach** - the committed delivery date has passed with no evidence of fulfillment. This Company cannot prove contractual compliance or assign operational responsibility for any of these cases.

3.3 Delivered orders with no invoice

```
SELECT 'SECTION 3 - Type C: Delivered orders with no invoice
(revenue leakage)' AS diagnostic_section;
```

```
SELECT
    p.pedido_id,
    p.numero_pedido,
    p.cliente_id,
    vc.nit,
    vc.nombre AS client_name,
    p.ruta_id,
    r.ciudad_origen + ' → ' + r.ciudad_destino AS route,
    e.fecha_entrega_real,
    e.estado_entrega,
    p.valor_declarado,
    p.peso_kg,
```

```

        p.tipo_carga,
        DATEDIFF(DAY, e.fecha_entrega_real, GETDATE()) AS days_de
livered_no_invoice
FROM operaciones.PEDIDO p
INNER JOIN operaciones.ENTREGA e
    ON p.pedido_id = e.pedido_id
    AND e.estado_entrega = 'ENTREGADO' -- Confirmed deli
very
LEFT JOIN facturacion.FACTURA f
    ON p.pedido_id = f.pedido_id -- No invoice for
this order
LEFT JOIN ventas.CLIENTE vc ON p.cliente_id = vc.cliente_id
LEFT JOIN operaciones.RUTA r ON p.ruta_id = r.ruta_id
WHERE f.factura_id IS NULL -- No invoice fou
nd
ORDER BY p.valor_declarado DESC ;

SELECT
    COUNT(*) AS delivered_never_invoic
ed,
    SUM(p.valor_declarado) AS revenue_leakage_total,
    AVG(p.valor_declarado) AS avg_uninvoiced_deliver
y,
    COUNT(DISTINCT p.cliente_id) AS clients_with_uninvoice
d_deliveries
FROM operaciones.PEDIDO p
INNER JOIN operaciones.ENTREGA e ON p.pedido_id = e.pedido_id
AND e.estado_entrega = 'ENTREGADO'
LEFT JOIN facturacion.FACTURA f ON p.pedido_id = f.pedido_i
d
WHERE f.factura_id IS NULL;

```

diagnostic_section											
1	SECTION 3 - Type C: Delivered orders with no invoice (revenue leakage)										
pedido_id	numero_pedido	cliente_id	nit	client_name	ruta_id	route	fecha_entrega_real	estado_entrega	valor_declarado	peso	
1	500564	PED-000499159	75	NIT-000075	Procesadora de Alimentos GT 75	2077	Zacapa ? Escuintla	2026-02-14 20:00:00.0000000	ENTREGADO	60992.00	21185
2	298973	PED-000297568	12679	NIT-012679	Servicios Integrados SA 12679	1010	Chimaltenango ? Sacatepequez	2026-02-15 05:00:00.0000000	ENTREGADO	36755.00	27157
3	332371	PED-000330966	8497	NIT-008497	Electrodomesticos del Sur 8497	748	Quetzaltenango ? Chiquimula	2026-02-14 17:00:00.0000000	ENTREGADO	9454.00	1863
delivered_never_invoiced	revenue_leakage_total	avg_uninvoiced_delivery	clients_with_uninvoiced_deliveries								
1	3	107201.00	35733.666666	3							

Query executed successfully. c13/lab_db (16.0 RTM) | carlos_roberto (52) | TRANSTRACK | 00:00:13 | Row: 1, Col: 1 | 1 rows

3 confirmed deliveries with no invoice, representing **\$107,201** in uncaptured revenue. While limited in this environment, the pattern demonstrates that the billing process has no automated reconciliation against completed deliveries — revenue leakage goes undetected without a governance control.

3.4 Duplicate invoices

```
SELECT 'SECTION 4 - Duplicate invoices (same pedido, two FACTURA records)' AS diagnostic_section;
```

```
SELECT
    f.pedido_id,
    COUNT(*) AS invoice_count,
    SUM(f.total) AS total_billed,
    MIN(f.total) AS min_invoice,
    MAX(f.total) AS max_invoice,
    MAX(f.total) - MIN(f.total) AS amount_discrepancy,
    STRING_AGG(
        CAST(f.factura_id AS VARCHAR) + ':' + f.numero_factura
        + '($' + CAST(f.total AS VARCHAR) + ')',
        ' | '
    ) AS invoice_details,
    MIN(f.fecha_emision) AS first_issued,
    MAX(f.fecha_emision) AS last_issued
FROM facturacion.FACTURA f
WHERE f.pedido_id IS NOT NULL
GROUP BY f.pedido_id
```

```
HAVING COUNT(*) > 1
ORDER BY total_billed DESC;
```

diagnostic_section									
SECTION 4 - Duplicate invoices (same pedido, two.									
pedido_id	invoice_count	total_billed	min_invoice	max_invoice	amount_discrepancy	invoice_details	first_issued	last_issued	
1 4748	2	122655.92	28000.00	94685.92	66685.92	489748 FAC-DUP-04748(\$28000.00) 477741 FAC-000	2026-01-31 00:00:00.0000000	2026-06-02 02:07:17.6900000	
2 4888	2	122020.47	28000.00	94020.47	66020.47	488898 FAC-DUP-04888(\$28000.00) 347440 FAC-000	2024-02-26 00:00:00.0000000	2026-06-02 02:07:16.8633333	
3 3188	2	121860.54	28000.00	93860.54	65860.54	488188 FAC-DUP-03188(\$28000.00) 242314 FAC-000	2022-08-03 00:00:00.0000000	2026-06-02 02:07:15.9066667	
4 1485	2	120904.06	28320.00	94584.06	63264.06	183762 FAC-000001485(\$94584.06) 486485 FAC-DUP	2021-11-18 00:00:00.0000000	2026-06-02 02:07:13.8333333	
5 4188	2	120755.26	28000.00	92755.26	64755.26	489188 FAC-DUP-04188(\$28000.00) 389704 FAC-000	2024-06-26 00:00:00.0000000	2026-06-02 02:07:17.0600000	
6 2394	2	120692.97	25760.00	94922.97	69162.97	326550 FAC-000002394(\$94922.97) 487384 FAC-DUP	2023-11-05 00:00:00.0000000	2026-06-02 02:07:15.0266667	
7 3544	2	120458.30	25760.00	94898.30	68938.30	488544 FAC-DUP-03544(\$25760.00) 318106 FAC-000	2023-09-19 00:00:00.0000000	2026-06-02 02:07:16.3233333	
8 4349	2	120392.78	28560.00	91832.78	63272.78	1975 FAC-000004349(\$91832.78) 489349 FAC-DUP-0	2019-01-12 00:00:00.0000000	2026-06-02 02:07:17.2400000	
9 3094	2	120376.42	25760.00	94616.42	68856.42	488094 FAC-DUP-03094(\$25760.00) 471010 FAC-000	2025-12-26 00:00:00.0000000	2026-06-02 02:07:15.8066667	
10 4143	2	120052.62	25200.00	94850.62	69650.62	489143 FAC-DUP-04143(\$25200.00) 154359 FAC-000	2021-04-14 00:00:00.0000000	2026-06-02 02:07:17.0000000	
11 3488	2	118819.45	28000.00	91819.45	63819.45	488498 FAC-DUP-03488(\$28000.00) 370781 FAC-000	2024-07-02 00:00:00.0000000	2026-06-02 02:07:16.2733333	
12 2593	2	118627.93	25200.00	94427.93	69227.93	70328 FAC-000002593(\$94427.93) 487583 FAC-DUP-	2020-01-17 00:00:00.0000000	2026-06-02 02:07:15.2966667	
13 3347	2	118560.28	27440.00	92120.28	64680.28	19454 FAC-000003347(\$92120.28) 488347 FAC-DUP-	2019-04-16 00:00:00.0000000	2026-06-02 02:07:16.0933333	
14 1446	2	118498.18	26880.00	92618.18	65738.18	486446 FAC-DUP-01446(\$26880.00) 289047 FAC-000	2023-04-15 00:00:00.0000000	2026-06-02 02:07:13.7633333	
15 4647	2	118759.65	27440.00	91319.65	63739.65	331868 FAC-000004647(\$91319.65) 489647 FAC-DUP	2023-12-03 00:00:00.0000000	2026-06-02 02:07:17.5966667	
16 1840	2	118715.24	23520.00	95195.24	71675.24	192466 FAC-000001840(\$95195.24) 486840 FAC-DUP	2021-11-06 00:00:00.0000000	2026-06-02 02:07:14.3133333	
17 4148	2	118635.16	28000.00	90635.16	62635.16	170288 FAC-000004148(\$90635.16) 489148 FAC-DUP	2021-07-09 00:00:00.0000000	2026-06-02 02:07:17.0066667	
18 4943	2	118617.86	25200.00	93417.86	66217.86	489943 FAC-DUP-04943(\$25200.00) 24990 FAC-0000	2019-05-16 00:00:00.0000000	2026-06-02 02:07:17.8966667	
19 3041	2	118557.43	24080.00	94477.43	70397.43	406760 FAC-000003041(\$94477.43) 488041 FAC-DUP	2025-01-13 00:00:00.0000000	2026-06-02 02:07:15.7466667	
20 4344	2	118238.33	25760.00	92478.33	67116.33	489344 FAC-DUP-04344(\$25760.00) 473189 FAC-000	2026-01-08 00:00:00.0000000	2026-06-02 02:07:17.2333333	
21 3948	2	117894.46	28000.00	89894.46	61894.46	488948 FAC-DUP-03948(\$28000.00) 57413 FAC-0000	2019-11-08 00:00:00.0000000	2026-06-02 02:07:16.7900000	
22 4793	2	117853.87	25200.00	92653.87	67463.87	250862 FAC-000004793(\$92653.87) 489783 FAC-DUP	2022-09-18 00:00:00.0000000	2026-06-02 02:07:17.7466667	
23 1008	2	112764.05	28000.00	80384.05	61264.05	488085 FAC-DUP-01008(\$28000.00) 204353 FAC-000	2023-01-11 00:00:00.0000000	2026-06-02 02:07:14.6193333	

Total_overbilled_amount

```
SELECT
    COUNT(*) AS pedidos_billed_twice,
    SUM(total_billed - min_invoice) AS overbilled_amount
FROM (
    SELECT
        pedido_id,
        SUM(total) AS total_billed,
        MIN(total) AS min_invoice
    FROM facturacion.FACTURA
    WHERE pedido_id IS NOT NULL
    GROUP BY pedido_id
    HAVING COUNT(*) > 1
) dup;
```

	min_discrepancy	max_discrepancy	avg_discrepancy
1	2.74	93280.10	35806.520463

3,452 orders were invoiced twice with discrepancies ranging from **\$2.74 to \$93,280** per case, averaging **\$35,806 per duplicate**. Total overbilling exposure: **\$170.2M** accumulated since 2019 - approximately **\$24M annually**. No constraint or reconciliation process exists to detect or prevent this at the system level.

3.5 Consolidated orphan summary for CTO report

```
SELECT 'SECTION 5 - Consolidated Orphan Summary' AS diagnosti
c_section;
```

```
SELECT orphan_type, record_count, financial_exposure
FROM (
    SELECT
        'Type A: Invoices with no order (NULL pedido_id)' AS
orphan_type,
```

```
        COUNT(*)          AS record_count,
        SUM(f.total)       AS financial_exposure
```

```
FROM facturacion.FACTURA f
WHERE f.pedido_id IS NULL
```

```
UNION ALL
```

```
SELECT
    'Type B: Orders with no delivery record',
    COUNT(*),
    SUM(p.valor_declarado)
FROM operaciones.PEDIDO p
LEFT JOIN operaciones.ENTREGA e ON p.pedido_id = e.pedido
_id
WHERE e.entrega_id IS NULL AND p.estado NOT IN ('CANCELAD
O')
```

```
UNION ALL
```

```

SELECT
    'Type C: Delivered, never invoiced (revenue leakag
e)',
    COUNT(*),
    SUM(p.valor_declarado)
FROM operaciones.PEDIDO p
    INNER JOIN operaciones.ENTREGA e ON p.pedido_id = e.pedid
o_id AND e.estado_entrega = 'ENTREGADO'
    LEFT JOIN facturacion.FACTURA f ON p.pedido_id = f.pedid
o_id
WHERE f.factura_id IS NULL

UNION ALL

SELECT
    'Type D: Duplicate invoices (same pedido billed twic
e)',
    SUM(invoice_count - 1), -- excess invoices
    SUM(excess_billed)
FROM (
    SELECT
        COUNT(*) AS invoice_count,
        SUM(total) - MIN(total) AS excess_billed
    FROM facturacion.FACTURA
    WHERE pedido_id IS NOT NULL
    GROUP BY pedido_id
    HAVING COUNT(*) > 1
) dup
) AS summary
ORDER BY financial_exposure DESC;

```

diagnostic_section

1SECTION 5 - Consolidated Orphan Summary

orphan_type	record_count	financial_exposure
1Type B: Orders with no delivery record	11799	599661971.00
2Type D: Duplicate invoices (same pedido billed tw...	3452	170287507.72
3Type A: Invoices with no order (NULL pedido_id)	5000	141608042.24
4Type C: Delivered, never invoiced (revenue leaka...	3	107201.00

Query executed successfully.

c:\3\lab_db (16.0 RTM)carlos_roberto (61)TRANSTRACK00:00:00Row: 1, Col: 14 rows